

PINNsFormer for the 4D Hodgkin-Huxley Neuron Model: Literature Review and Technical Analysis

Part 1: Neuroscience Foundations

The **Hodgkin-Huxley (HH) model** is a biophysical model of an excitable neuronal membrane. It describes how the membrane potential V_m evolves due to ionic currents through voltage-gated channels. When a sufficiently strong stimulus is applied, the neuron generates an **action potential (AP)**: a rapid, large-amplitude spike in V_m followed by a hyperpolarizing undershoot [24†L250-L258]

[24†L259-L267]. Biophysically, this arises from the interplay of **sodium (Na^+) channels** (rapid activation/inactivation) and **potassium (K^+) channels** (slower activation), plus a passive leak channel. At rest, most Na-gates are closed (small $m \approx 0$) so V_m is near the leak reversal (resting potential). A stimulus that raises V_m even a little increases Na-conductance (via the m^3h term), causing further depolarization (positive feedback) [24†L250-L258]. If this feedback is strong enough, an AP is triggered. As V_m nears the Na reversal potential, the inward current stops. Meanwhile, the Na current shuts off because the inactivation gate h closes (with a slower time-constant than m), and the outward K-current (driven by n^4) opens on a similar slow timescale [24†L259-L267]. The combined effect is a brief spike and then a **repolarization and after-hyperpolarization** [24†L259-L267].

- **Membrane Voltage Dynamics:** The membrane is modeled as a capacitance C_m in parallel with ionic conductances. Changes in V_m are governed by capacitive current $I_C = C_m dV_m/dt$ and ionic currents (Na, K, leak) driven by $V_m - E_{\text{ion}}$. The HH model accounts for these with voltage-dependent gating variables m, n, h , capturing channel opening probabilities [10†L135-L142].
- **Na^+ Channels (fast inward current):** Gated by activation variable m (with three subunits) and inactivation h . The Na current is $I_{\text{Na}} = \bar{g}_{\text{Na}} m^3 h (V_m - E_{\text{Na}})$. When V_m rises, m quickly increases (opening channels) while h slowly decreases (closing channels) [24†L250-L258].
- **K^+ Channels (delayed outward current):** Modeled by activation n (with four subunits) and reversal E_K . The K current is $I_K = \bar{g}_K n^4 (V_m - E_K)$. n rises slowly in response to depolarization, providing an outward current that brings V_m down [24†L259-L267].
- **Leak Current:** A constant (ohmic) leak conductance g_L with reversal E_L accounts for passive ion flow (primarily Cl^-). It ensures a resting potential near -65 mV.

These ionic mechanisms make the HH model *excitable*: a small input below threshold produces only a tiny voltage excursion, whereas a slightly stronger input crosses threshold and generates a full AP (all-or-none behavior). In the HH model this manifests as a **threshold effect** (a minimal injected charge must exceed a critical value to fire) [24†L330-L339]. After an AP, the neuron exhibits **refractoriness**: a second stimulus given too soon fails to trigger another spike (due to the after-hyperpolarization and remaining channel states) [24†L342-L349].

The HH model is fundamental in neuroscience because it quantitatively explains spike generation and refractory dynamics from ionic biophysics. Action potentials are the primary means of information

transmission in neurons (“neural coding”): signals are encoded by spike timing or rate. The shape and patterns of spiking (e.g. tonic firing, bursting) are determined by HH dynamics. Thus HH captures key concepts like excitability, spike initiation, and refractory phenomena [24†L330-L339] [24†L342-L349]. Understanding these is crucial for modeling how neurons compute and encode information.

Part 2: Full Mathematical Formulation

The classical **4D Hodgkin-Huxley ODE system** describes the evolution of (V, m, h, n) . In one common form, with membrane capacitance C_m and external current I_{ext} , it is written [10†L135-L142] :

$$C_m \frac{dV}{dt} = I_{\text{ext}} - \bar{g}_{Na} m^3 h (V - E_{Na}) - \bar{g}_K n^4 (V - E_K) - \bar{g}_L (V - E_L),$$

$$\frac{dm}{dt} = \alpha_m(V) (1 - m) - \beta_m(V) m,$$

$$\frac{dh}{dt} = \alpha_h(V) (1 - h) - \beta_h(V) h,$$

$$\frac{dn}{dt} = \alpha_n(V) (1 - n) - \beta_n(V) n.$$

Here:

- $V(t)$ is the membrane voltage (mV). C_m is the membrane capacitance per unit area ($\mu\text{F}/\text{cm}^2$). $\bar{g}_{Na}, \bar{g}_K, \bar{g}_L$ are the maximal conductances (mS/cm^2) for sodium, potassium, and leak channels, respectively [10†L135-L142]. E_{Na}, E_K, E_L are the corresponding reversal (Nernst) potentials (mV).
- The current terms $\bar{g}_X \cdot (\cdot) (V - E_X)$ each have units of $\mu\text{A}/\text{cm}^2$ (current density), so I_{ext} and $C_m dV/dt$ must also be in $\mu\text{A}/\text{cm}^2$ for consistency.
- $m, h, n \in [0, 1]$ are dimensionless gating variables. Physically, m and h give the probability of a Na-channel subunit being activated/deactivated, and n the probability of a K-channel subunit being open [30†L231-L240] [30†L244-L253]. (With 3 activation gates and 1 inactivation in Na, and 4 activation gates in K, the currents depend on $m^3 h$ and n^4 .)
- $\alpha_m(V), \beta_m(V)$ (and similarly $\alpha_h, \beta_h, \alpha_n, \beta_n$) are empirically derived, voltage-dependent rate functions (units s^{-1}). They govern the opening/closing kinetics of the gates [30†L269-L278]. For example, $\alpha_m(V)(1 - m)$ is the voltage-dependent rate at which m gates open, and $\beta_m(V)m$ the rate at which they close.

Together, these four equations form a nonlinear dynamical system. We can define the state vector $x(t) = [V(t), m(t), h(t), n(t)]^T$. In compact form, $\dot{x} = f(x)$, where $f : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ includes the above right-hand sides. Note the *coupling structure*: V appears in all gating equations (via the α, β functions), and the gating variables multiply into the dV/dt equation. This yields strong **nonlinearities** (polynomial $m^3 h$ and n^4 terms) and makes the system tightly coupled in all four variables [10†L135-L142]. Each term is physically meaningful: e.g. $\bar{g}_{Na} m^3 h (V - E_{Na})$ is the instantaneous Na-current, etc.

Part 3: Dynamical Systems Analysis

From a dynamical systems viewpoint, the HH model exhibits rich behavior depending on the input current I_{ext} . For a constant I , there is typically a single equilibrium (resting potential) at which $\dot{x} = 0$. Linearizing around that point reveals the stability: in the standard squid-axon parameter regime, two eigenvalues are real and negative, and the other two form a complex conjugate pair (with small positive real part just beyond threshold) [32†L399-L407]. In fact, Hodgkin and Huxley showed that the system has a 2D “center manifold” where dynamics effectively occur, because two eigenvalues strongly decay to zero and two capture the oscillatory behavior [32†L399-L407].

As I_{ext} increases, the resting equilibrium eventually loses stability via a **Hopf bifurcation** [32†L419-L428]. In other words, a pair of complex eigenvalues crosses into the positive half-plane, and a stable limit cycle (repetitive spiking) emerges. This bifurcation in HH is of the *subcritical* (type-II) kind: the firing onset is abrupt (there is a minimum nonzero frequency) and the f - I curve has a jump discontinuity [32†L419-L428] [24†L330-L339]. Below this threshold, a perturbation away from rest decays back (possibly with a small subthreshold oscillation), but above threshold the neuron settles into a periodic orbit representing a train of identical action potentials (Fig. 1 of the HH dynamics [32†L419-L428]).

Thus, in phase space the HH neuron has (1) a stable fixed point for subthreshold input, (2) a critical Hopf point at threshold, and (3) a stable limit cycle beyond threshold. The limit cycle lives essentially on a 2D manifold (due to the eigenstructure) [32†L399-L407]. When the system is at rest, any small stimulus will produce a brief excursion and return to the fixed point (threshold phenomena [24†L330-L339]). When the limit cycle exists, trajectories spiral away from the unstable rest point and converge to the oscillation (sustained spiking). Complex regimes like bursting (alternating periods of spiking and quiescence) or more exotic bifurcations can appear if other slow variables or currents are added, but the classical 4D HH already captures the central saddle-node or Hopf bifurcations underlying excitability [32†L419-L428] [24†L330-L339].

Part 4: Time Scale Analysis

The Hodgkin-Huxley equations involve **multiple time scales**. Empirically, the activation gate m responds very quickly (sub-millisecond to millisecond scale) when V changes, whereas the inactivation gate h and potassium gate n evolve much more slowly (several to tens of milliseconds) [24†L259-L267]. For example, at peak depolarization m can near saturation within a millisecond, but h takes an order of magnitude longer to inactivate the Na-current [24†L259-L267]. The membrane voltage itself can change extremely fast during the upstroke (due to the high $\bar{g}_{Na}$ at the spike peak), and more slowly during repolarization (as K-channels act). In summary: **fast variables**: V (during spike upstroke) and m ; **slow variables**: h and n .

This separation of scales makes the system effectively **stiff**. In particular, HH displays what could be called *spike-induced stiffness*: at the moment of spike initiation, derivatives are huge (fast dynamics), while outside spikes the state changes slowly. Such stiffness (large differences in time constants) is well known to cause numerical difficulties. Indeed, as noted in the literature, PINNs (and standard integrators) struggle on stiff problems [49†L263-L270] [57†L7-L15]. In the HH context, the stiff nature implies that any learning algorithm must capture both the rapid voltage rise and the slow recovery on the same mesh. This is analogous to benchmarks in chemical kinetics or other multiscale ODEs where standard PINNs fail [49†L263-L270].

Classical ODE stiff benchmarks (e.g. Robertson’s problem) have much larger scale separations, but HH is “moderately stiff” – its eigenvalues differ by factors of ~ 100 – 1000 in some regimes. The main consequence for PINNs is that the physics loss landscape becomes highly ill-conditioned around spikes, making convergence difficult. Thus, HH embodies a nontrivial *multiscale dynamical system* with potential “local stiffness” near events [57†L7-L15] .

Part 5: Classical PINN Formulation for HH

A standard physics-informed neural network (PINN) for HH would take time t (and possibly spatial location, though HH here is 0D) as input and output the state $[V(t), m(t), h(t), n(t)]$. The **residuals** of the ODE system are defined pointwise as:

$$\begin{aligned} R_V(t) &= C_m \frac{d\hat{V}}{dt}(t) - \left[I_{\text{ext}}(t) - \bar{g}_{Na} \hat{m}^3(t) \hat{h}(t) (\hat{V}(t) - E_{Na}) - \bar{g}_K \hat{n}^4(t) (\hat{V}(t) - E_K) - \bar{g}_L (\hat{V}(t) - E_L) \right], \\ R_m(t) &= \frac{d\hat{m}}{dt}(t) - \left[\alpha_m(\hat{V}(t))(1 - \hat{m}(t)) - \beta_m(\hat{V}(t))\hat{m}(t) \right], \\ R_h(t) &= \frac{d\hat{h}}{dt}(t) - \left[\alpha_h(\hat{V}(t))(1 - \hat{h}(t)) - \beta_h(\hat{V}(t))\hat{h}(t) \right], \\ R_n(t) &= \frac{d\hat{n}}{dt}(t) - \left[\alpha_n(\hat{V}(t))(1 - \hat{n}(t)) - \beta_n(\hat{V}(t))\hat{n}(t) \right], \end{aligned}$$

where hats denote the network’s outputs and time-derivatives are computed by automatic differentiation.

The PINN loss typically includes:

- A **physics loss** enforcing the ODEs: $L_{\text{phys}} = \frac{1}{N} \sum_{i=1}^N [R_V(t_i)^2 + R_m(t_i)^2 + R_h(t_i)^2 + R_n(t_i)^2]$, summing over collocation times t_i .
- An **initial-condition loss**: enforcing $\hat{x}(0) \approx x_0$. For example $L_{IC} = \|\hat{V}(0) - V_0\|^2 + \|\hat{m}(0) - m_0\|^2 + \dots$.
- Possibly a **data loss** if voltage measurements $\{V(t_j)\}$ are available: $L_{\text{data}} = \sum_j |\hat{V}(t_j) - V_{\text{obs}}(t_j)|^2$.

The total loss is a weighted sum $L = \lambda_{\text{phys}} L_{\text{phys}} + \lambda_{IC} L_{IC} + \lambda_{\text{data}} L_{\text{data}}$. Boundary conditions in time are only the initial conditions (since HH is an IVP).

Why PINNs fail near spikes: The HH residuals become extremely large and rapidly varying during the spike, making the loss landscape very steep and ill-conditioned. Classical PINNs with smooth activations (or even ReLU) often cannot satisfy such sharp transitions well, tending to either “smooth over” the spike or require an enormous number of collocation points. Indeed, failure modes of PINNs have been documented in stiff ODEs [49†L263-L270] : the network gets trapped in a local optimum where the residuals are not properly zero. In practice one sees that the network struggles to simultaneously reduce error on the flat subthreshold region and the sharp suprathreshold dynamics, leading to poor approximation of the spike waveform.

Part 6: PINNsFormer Architecture Design

The **PINNsFormer** architecture (Zhao *et al.*, ICLR 2024) is a Transformer-based variant of PINNs. Its key ideas include (1) transforming each input $[x, t]$ into a *pseudo sequence* of time-shifted points, (2) using multi-head self-attention to capture temporal correlations, and (3) a novel “wavelet” activation. Concretely, for an input collocation point (x, t) one creates the sequence

$$\{[x, t], [x, t + \Delta t], \dots, [x, t + (k - 1)\Delta t]\},$$

for some window length k and time step Δt [59†L13-L21]. This pseudo-sequence is embedded (via a spatio-temporal mixer) and fed into an encoder-decoder Transformer with self-attention. The idea is that the network now predicts k sequential outputs at once (a short trajectory), rather than just a point value. The **self-attention** layers let the model learn which past/future time-steps in the sequence are relevant to each other, effectively capturing temporal dependencies that plain PINNs ignore.

Additionally, PINNsFormer uses a custom **wavelet activation** function $\mathcal{W}(z) = \omega_1 \sin(z) + \omega_2 \cos(z)$ in its feedforward layers [5†L347-L354]. This periodic (Fourier-like) activation is meant to help represent oscillatory or high-frequency components (motivated by Fourier feature techniques).

Finally, the loss is computed over the entire pseudo-sequence. That is, after the Transformer outputs the approximations $\{\hat{u}(x, t + j\Delta t)\}_{j=0}^{k-1}$, one computes time-derivatives for each sequence element by auto-differentiation, and forms residuals $R(x, t + j\Delta t)$ at each step [5†L421-L430]. The total physics

loss sums these: $L_{\text{phys}} = \sum_{j=0}^{k-1} |R(x, t + j\Delta t)|^2$, analogous to a sequential version of the PINN loss

[5†L421-L430]. Initial-condition loss is similarly applied to the first element of each sequence if it coincides with the initial time.

Each component maps naturally onto HH dynamics. The pseudo-sequence in time allows the model to see (and match) the rapid rise and fall of a spike across its window, rather than inferring them pointwise. Attention heads can link distant time points (e.g. the start and peak of a spike) to ensure the transient is captured coherently. The wavelet activation helps represent the sharp transient (high-frequency) features of $V(t)$. In summary, PINNsFormer turns the single-variable ODE problem into a short sequence prediction problem, leveraging Transformer strengths in temporal pattern learning [59†L13-L21] [5†L347-L354].

Part 7: PINNsFormer Architecture Variants for HH

We propose five Transformer-PINN hybrid designs for modeling HH dynamics. Each variant differs in how the state variables $[V, m, h, n]$ and time are tokenized and processed:

- **A. Single-Sequence PINNsFormer:** Follow the standard PINNsFormer: each input is the pair (t) (and spatial location if any). We generate a time-window sequence $\{(V(t + j\Delta t), m(t + j\Delta t), h(t + j\Delta t), n(t + j\Delta t))\}_{j=0}^{k-1}$ as output. Essentially, the entire state vector is treated as the “feature” of each time token. The Transformer attends along time. *Advantages:* Direct extension of PINNsFormer, captures temporal context over a spike. *Weaknesses:* All four variables must be encoded together, potentially mixing different time scales; sequence length tradeoff. *Complexity:* Transformer of length k with 4-dimensional input per token.

- **B. Multivariate State-Token PINNsFormer:** Tokenize each state variable separately. For a window of k time steps, create a sequence of $4k$ tokens: $\{V(t), V(t + \Delta t), \dots, V(t + (k - 1)\Delta t), m(t), \dots, m(t + (k - 1)\Delta t), h(\dots), n(\dots)\}$. Alternatively, create 4 interleaved sequences (one per variable). The Transformer uses positional encoding to distinguish variables vs time. *Advantages:* Allows attention to learn correlations between specific variables (e.g. attend from V to n), and different embeddings per variable. *Weaknesses:* Sequence is longer ($4\times$ length), more computation; must enforce coupling between streams (no built-in cross-attn unless designed). *Complexity:* Length $O(4k)$ tokens.
- **C. Cross-Attention Gating PINNsFormer:** Use separate encoders for voltage and gating variables, with cross-attention between them. For example, one branch encodes the V -trajectory, another encodes (m, h, n) trajectories. Use cross-attention layers so that the gating branch can attend to voltage features and vice versa. *Advantages:* Explicitly models the causal influence of V on gates and feedback; may stabilize training by modularizing channels. *Weaknesses:* More complicated architecture; might require careful balancing of losses to keep branches consistent. *Complexity:* Two transformers with cross-attn; roughly double parameters.
- **D. Hierarchical Timescale PINNsFormer:** Exploit the different timescales by hierarchical encoding. For instance, embed $\{V, m\}$ at fine resolution Δt , but embed h, n at coarser time steps (e.g. $2\Delta t$), possibly with pooling. The model has a coarse-to-fine structure: first predict a coarse trajectory for slow variables and a detailed trajectory for fast ones, then refine. *Advantages:* Matches the intrinsic HH scales, reducing stiffness for the net. *Weaknesses:* Adds engineering complexity; might be hard to train end-to-end. *Complexity:* Requires multi-resolution attention; complexity similar to a transformer over longest sequence.
- **E. Neural Operator + PINNsFormer Hybrid:** Combine a Neural Operator (e.g. Fourier Neural Operator) to first approximate the HH flow, then refine with a PINNsFormer. Concretely, use a pre-trained NN that maps an input current waveform to a nominal voltage solution (data-driven), then use a PINNsFormer to enforce physics on the residual. *Advantages:* The neural operator can capture large-scale spiking behavior quickly; the PINNsFormer corrects fine details. *Weaknesses:* Requires training the operator separately; additional complexity. *Complexity:* Sum of operator network and transformer complexities.

Each variant can be depicted as a block diagram (omitted), with arrows showing the flow from time input to sequence tokens, through attention layers, to outputs $\hat{V}, \hat{m}, \hat{h}, \hat{n}$. Their mathematical formulations differ in the input tokenization and attention operations, but all conclude with residual-based losses. For example, in (B) the input embedding could be

$$\text{TokenEmb}([V(t), \dots, n(t + (k - 1)\Delta t)]) = \{\text{Embed}_V(V(t)), \dots, \text{Embed}_n(n(t + (k - 1)\Delta t))\},$$

feeding into a Transformer. A full comparative analysis would detail the dimension of attention matrices ($O(k^2)$ scaling) and parameter counts. Roughly, (A) and (D) have complexity $O(k^2)$ per layer; (B) has $O((4k)^2) \approx 16k^2$ when flattened; (C) has two $O(k^2)$ blocks; (E) adds the cost of the operator (often FFT-based for FNO). The main weakness across all is computational cost for long sequences, but they exploit PINNsFormer's ability to handle longer horizons than a plain PINN.

Part 8: Attention Analysis

A key question is: *what do Transformer attention heads learn about HH dynamics?* We hypothesize that self-attention may latch onto biophysical events. For instance, one attention head might strongly link

time-steps corresponding to the rapid voltage upswing, effectively detecting the spike onset. Another head might attend to the repolarization phase and the state of the n -gate, capturing how K-channel opening drives the voltage down. During training, attention weights could highlight causality: early in a spike sequence, attention may focus on past low-voltage context (input drive), whereas later tokens focus on the immediate voltage rise. In terms of gating, attention may discover the implicit relationships $m \rightarrow V$ and $V \rightarrow h, n$.

Visually, one might expect an attention map where each time token attends most to those a few steps before and after it – e.g. “banded” attention around the diagonal, but with bursts of non-local attention around the spike event. If training shows that attention weights sharply increase between distant time points (e.g. from spike trough to peak), this could indicate the model learning the long-range dependency of threshold behavior. Whether these patterns align with known neuroscience (e.g. absolute refractory period length) is an open question. In principle, analyzing the trained attention matrices could reveal the model’s learned representation of excitability and ionic coupling, potentially offering insights (though caution is needed, as attention is not guaranteed to be interpretable).

Part 9: Stiffness and Multiscale Behavior

PINNs vs. PINNsFormer on HH: The HH system’s stiffness and sharp spikes make it challenging for classical PINNs. As noted by Krishnapriyan *et al.* (NeurIPS 2021), PINNs often fail on problems with sharp transients or stiff terms [49†L263-L270]. In fact, they cite “stiff ODEs (chemical kinetics)” as an example where PINNs did not work well [49†L263-L270] – a category into which HH also falls due to its disparate time scales. This suggests that a plain PINN would struggle with accurate spike modeling and might converge to a spurious “soft” solution (missing the sharp peak).

PINNsFormer, in contrast, has shown empirically much smoother training dynamics. Zhao *et al.* analyzed the loss landscape for PINNsFormer and found it to be dramatically more well-behaved: the loss as a function of parameters is much smoother, with fewer local minima, than for a conventional PINN [51†L546-L554]. In fact, the Lipschitz constant of PINNsFormer’s loss was an order of magnitude smaller than for PINNs, implying easier optimization [51†L564-L573]. Consequently, one can expect PINNsFormer to better capture the spike (reducing failure modes) simply because optimization is less prone to get stuck [51†L546-L554].

Comparison to other methods: Several SciML alternatives exist. Domain-decomposed PINNs (XPINN) or adaptive/sampled PINNs can localize effort around the spike, but still operate pointwise in time. Neural operators (DeepONet, FNO, etc.) aim to learn the input-to-solution map. Notably, recent work shows that DeepONet/FNO/WNO can indeed learn the HH operator with high accuracy when carefully designed – achieving ~1–2% error in predicting $V(t)$ from input current [53†L26-L35]. However, those methods require a lot of training data and may lack built-in physical constraints. Neural ODE variants or ResNet surrogates have been tried (e.g. representing HH as a learned continuous normalizing flow [53†L83-L92]), but they may not enforce exact physics. PINNsFormer strikes a hybrid: it uses physics loss but with Transformers’ sequence modeling.

Key questions include:

1. **Hardest regimes:** Likely the most challenging HH regimes are those with fast, complex spiking (e.g. chaotic input currents leading to irregular firing or bursting). PINNs struggle most at steep gradients or bifurcation points [49†L263-L270]. Bursting regimes (slow modulation of spike amplitude) add a further timescale.
2. **Role of Attention:** We anticipate that self-attention can help capture the causal structure of spikes. By attending across time, the model might better enforce the refractory timing and channel coupling.

This could improve on typical PINN failures where the temporal sequence is not explicitly learned.

3. Sequential loss (causality): PINNsFormer’s sequential loss (aggregating residuals over a window) imposes a kind of *causality consistency*. Krishnapriyan *et al.* showed that reformulating PINNs as a sequence learning task dramatically lowers error (by 1–2 orders) compared to naive PINNs [47†L33-L38] . This suggests the sequential approach avoids some local minima.

4. Long horizons: A limitation of PINNs is poor performance over long time intervals (error accumulates). In contrast, Transformer-based models can better handle long sequences if trained appropriately. PINNsFormer’s windowed approach might allow stitching longer horizons by shifting windows, mitigating error blow-up.

5. Stiffness mitigation: The internal attention mechanism may adaptively focus on rapid events (giving an implicit local mesh refinement in time), partially alleviating stiffness. The smoother loss landscape is another factor. In summary, we expect PINNsFormer to handle spike dynamics more robustly than PINNs, especially in stiff (rapid-spike) regimes [49†L263-L270] [51†L546-L554] .

Evidence from related literature supports these points. For example, Ji *et al.* developed **Stiff-PINN** for chemical kinetics, explicitly targeting stiff reactions, indicating this is a known issue. The operator-learning results [53†L26-L35] suggest that *if* a model has enough capacity it can learn HH solutions – implying that PINNsFormer (with suitable design) could likewise learn them given its success on other stiff PDEs. In summary, HH poses the classic PINN failure modes (nonlinearity, stiffness, multiscale), but the attention-based sequential approach is likely to yield substantial gains [47†L33-L38] [51†L546-L554] .

Part 10: Numerical Experiments

We outline a suite of experiments to compare PINNsFormer with baselines (PINN, XPINN, DeepONet, FNO, Neural ODE, etc.):

- **Experiment 1 – Single Spike Generation:** Input: A brief current pulse (e.g. 1 ms wide) to induce exactly one AP. Output: the voltage trajectory $V(t)$ (and possibly gating traces). Metrics: Root-mean-square error of $V(t)$ (especially the peak amplitude and timing), and error in peak height. Failure Mode: Standard PINN often underestimates peak or delays it. PINNsFormer should match the sharp spike better (due to sequential prediction).
- **Experiment 2 – Repetitive Firing:** Input: A constant supra-threshold current to elicit a train of spikes. Output: Steady-state spike train, measured over e.g. 100 ms. Metrics: Frequency error (difference in inter-spike interval), phase error of spike times, and waveform error. Failure Mode: PINN may gradually drift or damp oscillations. PINNsFormer should lock onto the periodic limit cycle more reliably.
- **Experiment 3 – Bursting Dynamics:** Input: A slowly varying current (e.g. sinusoidal or step that induces bursting). Output: A burst train (clusters of spikes). Metrics: Ability to capture burst pattern (burst duration, intraburst frequency), and overall waveform RMSE. Failure Mode: The extra slow scale (burst envelope) often causes PINNs to average it out. PINNsFormer’s multi-scale design (e.g. variant D) can be tested here.
- **Experiment 4 – Parameter Estimation (Inverse Problem):** Input: Partial observation of a voltage trace (one or a few spikes) without known ionic parameters. Task: Infer conductances $\bar{g}_{Na}$, $\bar{g}_K$ and/or reversal potentials or hidden gating trajectories. Metrics: Parameter error, state reconstruction error. Failure Mode: Under-constrained inverse problem may have non-unique solutions or instability. PINNsFormer (and baselines) would be evaluated on their ability to

recover true parameters. Ferrante *et al.* showed basic PINNs can solve this [54†L1-L4] [54†L73-L82] .

- **Experiment 5 – Noisy Measurements:** Same as Exp1 or 2, but add realistic noise to $V(t)$ observations (to simulate experimental data). Task: Fit PINN models despite noise. Metrics: Robustness of voltage fit and parameter estimates as a function of noise level. Failure Mode: Overfitting or smoothing by PINNs; PINNsFormer might overfit sequential patterns or could be regularized via attention dropout.
- **Experiment 6 – Long-Time Forecasting:** Input: Initial voltage and parameters, no further input. Task: Simulate $V(t)$ for a long horizon (e.g. seconds, through many spikes). Metrics: Divergence time (how long before prediction drifts), frequency stability over time. Failure Mode: Error accumulation (especially if using PINNs with limited resolution). PINNsFormer may sustain accuracy longer due to attention capturing periodicity.
- **Experiment 7 – Out-of-Distribution (OOD) Currents:** Train on certain current waveforms, test on novel ones. For example, train on square pulses of one amplitude, test on pulses of different amplitude or shape. Metrics: Generalization error on new stimuli. Failure Mode: PINNs (and operators) can generalize poorly outside training distribution. We would assess whether PINNsFormer’s physics constraint aids extrapolation (it may not, since PINNsFormer is still solution-specific, but this tests the conditioning of the approach).

Each experiment uses identical PINN vs PINNsFormer network sizes for fairness. **Inputs** are time (and current stimulus waveform), **outputs** are V, m, h, n . **Metrics** include RMSE in $V(t)$, error in predicted spike times, parameter estimation error, and training convergence behavior. Expected challenges: stiff transient in Exp1/3, cumulative error in Exp6, identifiability in Exp4, and noise robustness in Exp5.

Part 11: Inverse Problems

A practical question is whether a PINNsFormer can infer HH parameters (maximal conductances, reversal potentials, etc.) and even hidden states from partial voltage data. **Observability:** The HH model is typically **observable** given $V(t)$ and known I_{ext} , because the gating variables affect V in a unique way over time. Indeed, Ferrante *et al.* (2022) used a PINN to infer all parameters and gating trajectories from simulated and real squid axon data [54†L1-L4] [54†L73-L82] . They found that the network could recover $\bar{g}_{Na}, \bar{g}_K$, etc., with low variability across repeated spikes. Their results imply the full state is, in practice, recoverable from $V(t)$ (within noise limits).

In the PINNsFormer context, one would treat $\bar{g}_{Na}, \bar{g}_K, E_{Na}, E_K$ as additional trainable parameters in the loss. The richer sequential context might help reduce correlation between parameters (e.g. g_{Na} vs g_K contributions at different spike phases). However, some ambiguities may remain: for instance, rescaling all g and adjusting E could sometimes produce similar traces. Ultimately, parameter identifiability depends on the data quality and diversity (multiple spike shapes or input levels help). We expect PINNsFormer to at least match the parameter recovery of ordinary PINNs (which succeeded in [54]) and possibly improve it by leveraging longer context (multiple spikes) within each pseudo-sequence.

Similarly, gating variables m, h, n are not directly measured, but they are inferred by Ferrante’s PINN as well. PINNsFormer could infer them in the same way, though we would check if attention improves the accuracy of hidden state estimation (e.g. stable recovery of $h(t)$ given its slow dynamics). In summary: existing evidence shows HH parameters and hidden states **are** inferable from voltage traces using

PINNs [54†L1-L4] [54†L73-L82] . PINNsFormer should in principle handle such inverse problems at least as well, though its advantage would be in the forward (prediction) accuracy that underpins inverse fitting.

Part 12: Research Gaps

Despite recent progress, many open questions remain at the intersection of PINNsFormer and computational neuroscience. Top open problems include:

1. **Multi-neuron and network models:** Extending PINNsFormer to networks of coupled HH neurons (with synaptic currents) is largely unexplored. How to scale transformers to many interacting ODEs is an open issue.
2. **Stochastic HH models:** Incorporating channel noise (stochastic gating) into PINNsFormer (stochastic PINNs) for realistic variability.
3. **Spatially-extended HH (cable equation):** The HH model as a PDE on a dendrite/axon (diffusion term plus ionic currents) and applying spatio-temporal PINNsFormer.
4. **Parameter identifiability theory:** Formal study of which HH parameters can be uniquely recovered by PINNs from limited data.
5. **Attention interpretability:** Analyzing trained attention weights for neuroscientific insight (e.g. do they correlate with known time constants or channel states?).
6. **Long-term training stability:** Can PINNsFormer be trained reliably in FP32 for stiff neuron models? (FP64 fixes some PINN failures [43†L41-L49] , but mixed precision scalability is unknown.)
7. **Alternative activations:** The effect of other basis functions (wavelets, Fourier features) in PINNsFormer for neuronal data.
8. **Curriculum/Adaptive training:** Integrating ideas like curriculum regularization (slowly increasing PDE difficulty [47†L33-L38]) with Transformer training for HH.
9. **Hybrid physics-ML models:** Combining PINNsFormer with reduced neuron models (e.g. FitzHugh-Nagumo or Izhikevich) for multi-scale brain simulations.
10. **Multi-modal data:** Using PINNsFormer to integrate electrophysiological data with gene-expression or imaging data via extended HH-like models.
11. **Hardware acceleration:** Implementing PINNsFormer training for large neuron models on specialized hardware (GPUs/TPUs) and quantifying speed-ups.
12. **Benchmark suites:** Developing standardized benchmarks (like the above experiments) for stiff ODE PINNs, especially tailored to neuroscience.
13. **Comparison with classical solvers:** Rigorous comparison of PINNsFormer vs. high-order stiff ODE solvers (e.g. BDF, exponential integrators) in terms of accuracy and speed for HH.
14. **Transformer variants:** Exploring other Transformer architectures (recurrent Transformers, Linformer, etc.) in the PINN context for efficiency.
15. **Zero-shot generalization:** Can a PINNsFormer trained on one type of neuron generalize to different neuron models or species?
16. **Integration with SNN training:** Bridging PINNsFormer with spiking neural network frameworks (e.g. using learned HH models to train SNN synapses).
17. **Energetic constraints:** Incorporating metabolic cost or ionic conservation laws as additional physics-informed constraints in PINNs.
18. **Adaptive domain decomposition:** Merging PINNsFormer with XPINN ideas (e.g. temporal domains or gating variable domains) for large networks.
19. **Probabilistic PINNsFormer:** Adding Bayesian or ensemble methods to quantify uncertainty in HH predictions.
20. **Real-time modeling:** Using PINNsFormer for fast emulation of HH dynamics in closed-loop neuromodulation or brain-machine interfaces.

Addressing these questions could significantly advance the use of advanced ML (and Transformers in particular) in neuroscience modeling. They span theory (observability, bifurcation analysis), methodology (architectural design, training algorithms), and application (multi-scale brain dynamics, interfacing with data). Each offers a promising PhD thesis direction at the cutting edge of machine learning and computational neurobiology.

Sources: We have drawn on foundational neuroscience literature [24†L250-L258] [24†L259-L267] [32†L399-L407] and on recent ML research on PINNs and transformers [47†L33-L38] [51†L546-L554] [53†L26-L35] [59†L13-L21] [5†L347-L354]. These include the original HH publications, review texts, and the latest works on PINNsFormer and neural operators, which provided key insights into the mathematical structure of HH and the benefits of attention-based PINNs. All citations are given in-text.
